

Gamma and Exponential Distribution

- Exponential distribution is a special case of Gamma distribution.
- Gamma and Exponentially distribution is a distributions that arises naturally in processes for which the waiting times between events are relevant. It can be thought of as a waiting time between Poisson distributed events.

Gamma and Exponential Distribution

- To predict the wait time until future events.
- The exponential distribution predicts the wait time until the ***very first*** event.
- The gamma distribution, on the other hand, predicts the wait time until the *** α -th*** event occurs.

Gamma Distribution

- A continuous random variable X follows gamma distribution with parameter α and β then its probability density function is given by

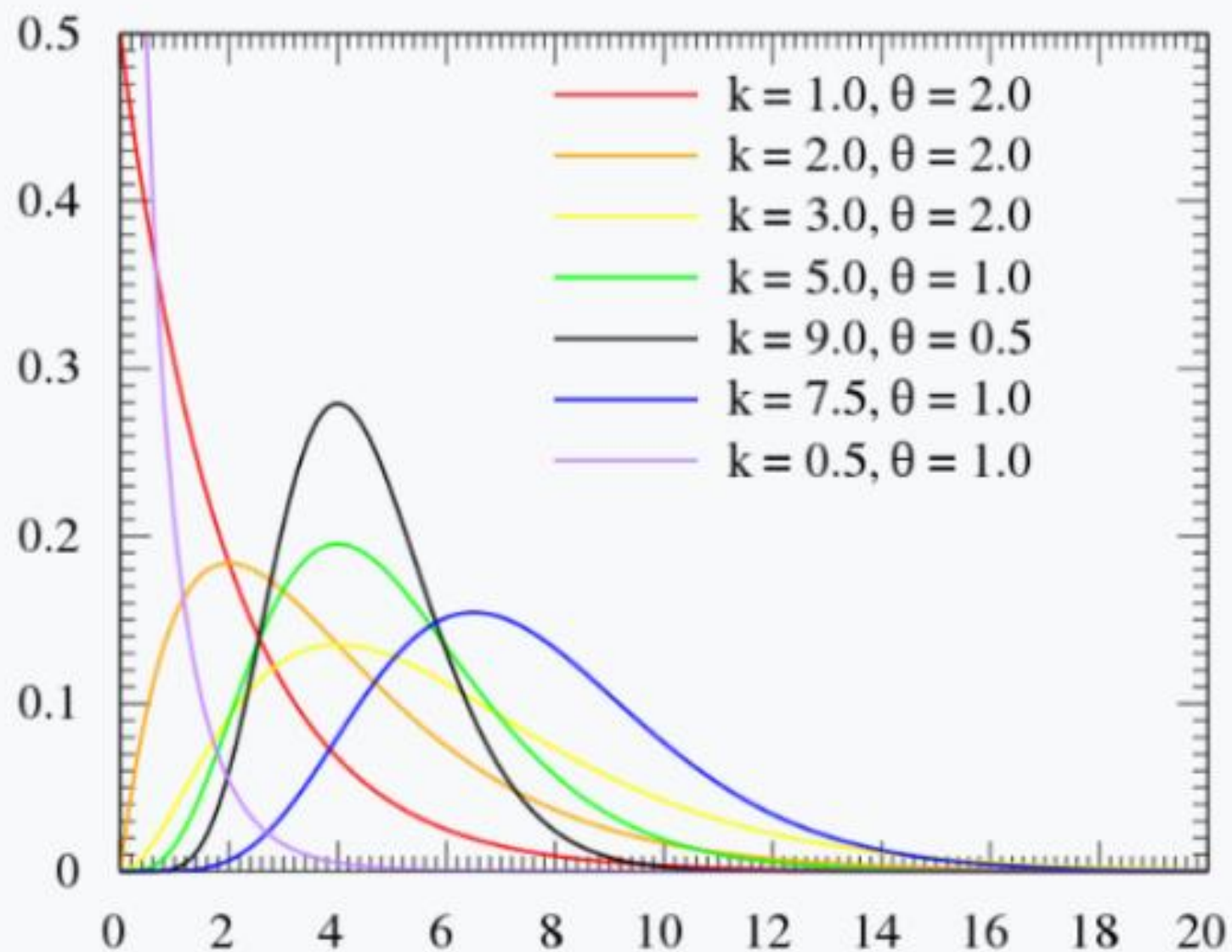
$$f(x, \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-\frac{1}{\beta}x}, & x > 0 \\ 0, & \textit{otherwise} \end{cases}$$

- where $\alpha > 0, \beta > 0$
- $X \sim \textit{Gamma}(\alpha, \beta)$.

Gamma Distribution

- $\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$: gamma function. $\alpha > 0$. $\alpha \in \mathbb{R}$.
- $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$, $\alpha > 0$, $\alpha \in \mathbb{R}$.
- $\int_0^{\infty} x^{\alpha-1} e^{-\lambda x} dx = \frac{\Gamma(\alpha)}{\lambda^{\alpha}}$, for $\lambda > 0$
- Note $\Gamma(n) = (n - 1)\Gamma(n - 1) = (n - 1)!$, n is integer.
- $\Gamma(1) = 1$
- $\Gamma\left(\frac{1}{2}\right) = \int_0^{\infty} \frac{1}{\sqrt{x}} e^{-x} dx = \sqrt{\pi}$
- $\Gamma\left(\frac{7}{2}\right) = \frac{5}{2} * \Gamma\left(\frac{5}{2}\right) = \frac{5}{2} * \frac{3}{2} * \Gamma\left(\frac{3}{2}\right) = \frac{5}{2} * \frac{3}{2} * \frac{1}{2} * \Gamma\left(\frac{1}{2}\right) = \frac{15}{8} \sqrt{\pi}$

Probability density function



Exponential Distribution $\sim$ Gamma($\alpha = 1, \beta$)

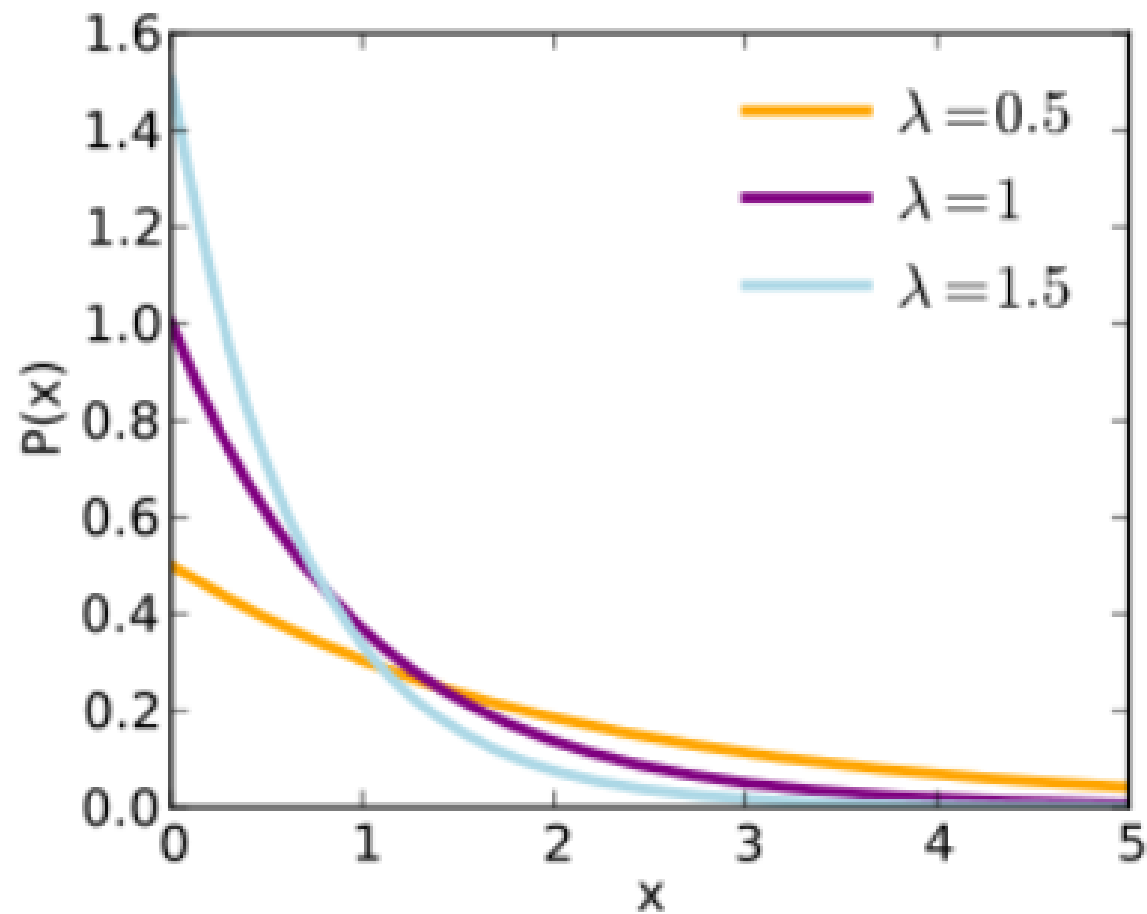
- A continuous random variable X follows exponential distribution with parameter β then its probability density function is given by

$$f(x, \beta) = \begin{cases} \frac{1}{\beta} e^{-\frac{1}{\beta}x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

- where $\beta > 0$: (some book they use $\frac{1}{\beta} = \lambda$)

Exponential Distribution

- To get some intuition for this interpretation of the exponential distribution, suppose you are waiting for an event to happen. For example, you are at a store and are waiting for the next customer. In each millisecond, the probability that a new customer enters the store is very small. You can imagine that, in each millisecond, a coin is tossed, and if it lands heads a new customer enters. If you toss a coin every millisecond, the time until a new customer arrives approximately follows an exponential distribution.



Probability Density Function of Exponential Distribution

Properties

Gamma Distribution

- The expected value :
 $E(x) = \alpha\beta.$
- The variance :
 $var(x) = \alpha\beta^2.$
- The moment generating function :
$$M_X(t) = E(e^{tX})$$
$$= (1 - \beta t)^{-\alpha}, t < \frac{1}{\beta}$$

Exponential Distribution

- The expected value :
 $E(x) = \beta.$
- The variance :
 $var(x) = \beta^2.$
- The moment generating function : $M_X(t) =$
$$E(e^{tX}) = (1 - \beta t)^{-1}, t < \frac{1}{\beta}$$

Exponential(memory less) distribution

- Think of an exponential random variable in the sense of tossing a lot of coins until observing the first heads. If we toss the coin several times and do not observe a heads, from now on it is like we start all over again. In other words, the failed coin tosses do not impact the distribution of waiting time from now on. The reason for this is that the coin tosses are independent. We can state this formally as follows: $P\left(\frac{X > x+a}{X > a}\right) = P(X > x)$. This is called a memory less property.

Memory less property

- $P\left(\frac{X > x+a}{X > a}\right) = P(X > x)$
- $\{ \text{WKT, } P(X > k) = \int_0^\infty \frac{1}{\beta} e^{-\frac{1}{\beta}x} dx = e^{-\frac{k}{\beta}} \}$
- $P\left(\frac{X > x+a}{X > a}\right) = \frac{e^{-\frac{x+a}{\beta}}}{e^{-\frac{a}{\beta}}} = e^{-\frac{x}{\beta}} = P(X > x)$
-

CDF of exponential distribution

- $P(X \leq x) = 1 - P(X > x)$
- $\{ \text{WKT}, P(X > k) = \int_0^\infty \frac{1}{\beta} e^{-\frac{1}{\beta}x} dx = e^{-\frac{k}{\beta}} \}$
- $P(X \leq x) = 1 - e^{-\frac{k}{\beta}}$
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- **Problem 1:** In a biomedical study with rats a dose response investigation is used to determine the effect of the dose of a toxicant on their survival time. The toxicant is a poison discharged into the atmosphere from jet fuel. For a certain dose of the toxicant the study determines the survival time (in weeks) has a gamma distribution with $\beta=10$ and $\alpha=5$. What is the probability that a rat survives not more than 60 weeks?

- Ans 1:

- X : survival time of a rat(in weeks)

- $X \sim \text{Gamma}(\alpha = 5, \beta = 10)$.

- The Pdf of X is $f(x) = \frac{1}{10^5 \Gamma(5)} x^{5-1} e^{-\frac{1}{10}x} ; x > 0$

- The Pdf of X is $f(x) = \frac{x^4 e^{-\frac{x}{10}}}{10^5 * 4!} ; x > 0$

Q1:

- P(probability that a rat survives not more than 60 weeks) = $P(X < 60)$

- $P(X < 60) = \int_0^{60} \frac{x^4 e^{-\frac{x}{10}}}{10^5 * 4!} dx = 0.715.$

- **Problem 2:** In a certain city, the daily consumption of electric power, in millions of kilowatt-hours, is a random variable X having a gamma distribution with mean 6 and variance 12.
 - (a) Find the values of α and β .
 - (b) Find the probability that on any given day the daily power consumption will exceed 12 million kilowatt-hours.

- **Ans 2:**

- X : The daily consumption of electric power, in millions of kilowatt-hours
- $X \sim \text{Gamma}(\alpha, \beta)$.
- Given mean = $\alpha\beta = 6$ and variance $\alpha\beta^2 = 12$.
- Q1 : To find α and β .
- Solve the above two equations, $\alpha = 3, \beta = 2$

- The Pdf of X is $f(x) = \frac{1}{2^3 \Gamma(3)} x^{3-1} e^{-\frac{1}{2}x} ; x > 0$
- Q2 : $P(X > 12)$
- $P(X > 12) = \int_{12}^{\infty} \frac{x^2 e^{-\frac{x}{2}}}{2^3 * 2!} dx = 0.0620$

- **Problem 3:** The amount of time that a watch will run without having to be reset is a random variable having an exponential distribution with mean of 120 days. Find the probability that such a watch will (a) have to be reset in less than 24 days. (b) not have to reset at least 180 days.

- Ans 2:

- X :is the amount of time a watch will run without being reset.
- $X \sim \text{Exp}(\beta)$. Here mean of X : $\beta = 120$.
- The Pdf of X is $f(x) = \frac{1}{\beta} e^{-\frac{1}{\beta}x} ; x > 0$
- The Pdf of X is $f(x) = \frac{1}{120} e^{-\frac{1}{120}x} ; x > 0$

- (a) : probability that it has to be reset in less than 24 days :
 $P(X < 24)$
- $P(X < 24) = \int_0^{24} \frac{e^{-\frac{x}{120}}}{120} dx = 0.18126$
- (a) : probability that it is not have to reset at least 180 days
: $P(X > 180)$
- $P(X > 180) = \int_{180}^{\infty} \frac{e^{-\frac{x}{120}}}{120} dx = 0.22313$

- **Problem 4:** The life, in years, of a certain type of electrical switch has an exponential distribution with an average life $\beta = 2$. If 100 of these switches are installed in different systems, what is the probability that at most 30 fail during the first year?

- Let random variable X represent the number of years that the switch survived.

X follows an exponential distribution with a mean $\beta = 2$.

-The probability density function of exponential distribution is:

$$f(x) = \frac{1}{\beta} e^{-\left(\frac{x}{\beta}\right)} \quad X > 0$$

- The probability distribution function of the exponential distribution is:

$$\begin{aligned} P(X \leq x) &= 1 - e^{-\left(\frac{x}{\beta}\right)} \\ &= 1 - e^{-\frac{x}{2}} \quad x > 0 \end{aligned}$$

First, let's find the probability that the number of switches fail during the first year.

$$\begin{aligned}P(X < 1) &= 1 - e^{-\frac{x}{2}} \\&= 1 - e^{-\frac{1}{2}} \\&= 1 - e^{-0.5} \\&= 1 - 0.6065 \\&= 0.3935\end{aligned}$$

- The total number of electrical switches $n = 100$.
- We consider it success if a switch fails during the first year.
- The probability of success is $p = 0.3935$.
- Hence, the probability of failure is $q = 1 - p = 1 - 0.3935 = 0.6065$

• Now, let random variable Y represent the number of switches failed during the first year.

Therefore, X has binomial distribution with parameters $n = 100$ and $p = 0.3935$.

Lets find the probability that at most 30 fail during the first year.

Because the sample size is very large, we have to use the normal approximation to binomial distribution.

$$\begin{aligned}
 P(X \leq 30) &= P(X \leq 30.5) \\
 &= P\left(\frac{X - np}{\sqrt{npq}} \leq \frac{30.5 - np}{\sqrt{npq}}\right) \\
 &= P\left(Z \leq \frac{30.5 - (100 \cdot 0.3935)}{\sqrt{100 \cdot 0.3935 \cdot 0.6065}}\right) \\
 &= P\left(Z \leq \frac{30.5 - 39.35}{\sqrt{23.8658}}\right) \\
 &= P(Z \leq -1.81) \\
 &= 0.0351
 \end{aligned}$$

- **Problem 4:** In a certain city, the daily consumption of water (in millions of liters) follows approximately a gamma distribution with $\alpha = 2$ and $\beta = 3$. If the daily capacity of that city is 9 million liters of water, what is the probability that on any given day the water supply is inadequate?

-The daily capacity of city is 9 million liters of water.

- Let random variable X represent the daily consumption of water.

- X follows approximately a gamma distribution with parameters $\alpha = 2$ and $\beta = 3$.

Therefore, the probability density function of gamma distribution is

$$f(x; 2, 3) = \begin{cases} \frac{1}{(3)^2 \Gamma(2)} x^{2-1} e^{-x/3} & x > 0 \\ 0 & \text{Otherwise} \end{cases}$$

If the daily capacity of city is 9 million liters of water, lets calculate the probability that on any given day the water supply is inadequate is.

$$\begin{aligned}P(X > 9) &= 1 - P(X \leq 9) \\&= 1 - \int_0^9 f(x; 2, 3)dx \\&= 1 - \int_0^9 \frac{1}{9}xe^{-x/3}dx \\&= 1 - \frac{1}{9} \left[x \left(-3e^{-x/3} \right) - \int \left(-3e^{-x/3} \right) dx \right]_0^9 \\&= 1 - \frac{1}{9} \left[\left(-3xe^{-x/3} \right) - 9e^{-x/3} \right]_0^9 \\&= 1 - \left(-\frac{1}{3}xe^{-x/3} - e^{-x/3} \right)_0^9 \\&= 1 - \left(\left(-\frac{1}{3}(9)e^{-9/3} - e^{-9/3} \right) - \left(-\frac{1}{3}(0)e^{-0/3} - e^{-0/3} \right) \right) \\&= 1 - \left(\left(-3e^{-9/3} - e^{-9/3} \right) - (0 - 1) \right) \\&= 1 - (1 - 4e^{-3}) \\&= 1 - (1 - 0.1991) \\&= 1 - 0.8009 \\&= 0.1991\end{aligned}$$